CFD 2025-26 – Lab 10 – solution

Exercise 1

The implementation of the full solver is provided in `CFDlab10.ipynb`.

1. Denoting by $\boldsymbol{u}_{\text{in}}$ the inlet velocity and by $\Gamma_{\text{w}} = \partial\Omega \setminus (\Gamma_{\text{in}} \cup \Gamma_{\text{out}}$ the lateral walls, we define the following spaces:

$$V = \{\boldsymbol{v} \in [H^1(\Omega)]^2 \colon \boldsymbol{v} = \boldsymbol{u}_{\text{in}} \text{ on } \Gamma_{\text{in}}, \boldsymbol{v} = \boldsymbol{0} \text{ on } \Gamma_{\text{w}}\},$$
$$V_0 = \in [H^1_{\partial\Omega\setminus\Gamma_{\text{out}}}(\Omega)]^2, \qquad Q = H^1_{\Gamma_{\text{out}}}(\Omega),$$

where we remark that the pressure space Q is a subspace of $\underline{H^1(\Omega)}$: this is required for step 2 of the Chorin-Temam scheme to be well-posed.

Introducing proper FE spaces $V_h \subset V$, $V_{0,h} \subset V_0$, $Q_h \subset Q$, and a discrete lifting $\mathcal{R}_h \boldsymbol{u}_{\text{in}} \in V_h$ of the inflow data, and applying the FEM, we can obtain the fully discrete numerical method: [1]
Given $\boldsymbol{u}_h^0 = \boldsymbol{u}_{\text{Stokes},h}$, for each $n = 0, 1, \dots$ perform the following steps:

1) Find $\widetilde{\boldsymbol{u}}_h = \check{\widetilde{\boldsymbol{u}}}_h + \mathcal{R}_h \boldsymbol{u}_{\text{in}}$, with $\check{\widetilde{\boldsymbol{u}}}_h \in V_{0,h}$ such that, $\forall \boldsymbol{v}_h \in V_{0,h}$,

$$\begin{aligned}\left(\frac{\check{\widetilde{\boldsymbol{u}}}_h}{\Delta t} + (\boldsymbol{u}_h^n \cdot \nabla)\check{\widetilde{\boldsymbol{u}}}_h, \boldsymbol{v}_h\right) &+ \frac{2}{\text{Re}}(\varepsilon(\check{\widetilde{\boldsymbol{u}}}_h), \varepsilon(\boldsymbol{v}_h)) \\ = \left(\frac{\boldsymbol{u}_h^n}{\Delta t}, \boldsymbol{v}_h\right) &- (\nabla p_h^n, \boldsymbol{v}_h) + \int_{\Gamma_{\text{out}}} h_{\text{out}} \boldsymbol{n} \cdot \boldsymbol{v}_h \, ds \\ &- ((\boldsymbol{u}_h^n \cdot \nabla)\mathcal{R}_h \boldsymbol{u}_{\text{in}}, \boldsymbol{v}_h) - \frac{2}{\text{Re}}(\varepsilon(\mathcal{R}_h \boldsymbol{u}_{\text{in}}), \varepsilon(\boldsymbol{v}_h))\end{aligned}$$

2) Find $\delta p_h \in Q_h$ such that, $\forall q_h \in Q_h$,

$$(\nabla \delta p_h, \nabla q_h) = \left(-\frac{1}{\Delta t} \operatorname{div} \widetilde{\boldsymbol{u}}_h, q_h\right).$$

3) Project velocity: find $\boldsymbol{u}_h^{n+1}$ such that

$$(\boldsymbol{u}_h^{n+1}, \boldsymbol{v}_h) = (\widetilde{\boldsymbol{u}}_h - \Delta t \, \nabla \delta p_h, \boldsymbol{v}_h) \qquad \forall \boldsymbol{v}_h \in V_{0,h}.$$

4) Update pressure:

$$p_h^{n+1} = p_h^n + \delta p_h.$$

We point out that step 3 is indeed a projection step, in the L^2 scalar product, of the corrected velocity onto the space V_h. Indeed, the (incremental) pressure gradient $\nabla \delta p_h$

[1] Notice that $\frac{1}{\text{Re}}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T) = \frac{2}{\text{Re}}\varepsilon(\boldsymbol{u})$.

does not belong to V_h, thus the quantity $\widetilde{\boldsymbol{u}}_h - \Delta t\, \nabla \delta p_h$ is not in V_h either, and needs to be projected.

In the implementation, we keep the velocity and pressure spaces separate, since they are used to define different problems, and so we do also with the boundary conditions list. An exception to this separation is the initialization Stokes problem, in which we assemble the monolithic system to have a quick implementation. The forms associated to steps 1, 2, and 3 are implemented in the functions `predictor_forms`, `pressure_forms`, and `projection_forms`, respectively. For each of those steps we create a `LinearVariationalProblem` and a `LinearVariationalSolver`, that is then used in the time loop. Notice that in the definition of the problems and solvers, the implementation choices ensure that the data of each problem is automatically updated at each time step, as long as `u_old` and `p_old` are properly updated.

Solving the problem, we obtain a flow whose final state at $t = 50$ is displayed in Figure 1. The low value of Re allows the flow to be split almost symmetrically by the obstacle in the early part of the time evolution. After some time, however, this symmetry is broken in the wake, and vortex shedding occurs.

We can notice some (small) oscillations of the pressure field near the outlet: this is typical of the differential Chorin-Temam scheme, which introduces a splitting error that is known to determine a boundary layer in the pressure field.

2 The forces at each time can be computed as follows:

```
drag_vec[ii] = assemble(
    inner( ph_old*normal - 2.0/Re*dot(sym(grad(uh_old)),
        normal), Constant((1.,0.)) ) * ds(5) )
  )
lift_vec[ii]= assemble(
    inner( ph_old*normal - 2.0/Re*dot(sym(grad(uh_old)),
    normal), Constant((0.,1.)) ) * ds(5) )
  )
```

resulting in the time-dependent plot reported in Figure 2.

The drag is the force exerted on the object in the direction of the flow, thus it is positive (directed as $+\boldsymbol{i}$) and it is much larger than the lift force, which acts in the transverse direction. The low value of the lift force is due to the symmetry of the obstacle and the domain, for which very small average transversal force is expected (at least for small Re). The vortex shedding developing in the wake can be seen in the physical oscillations of the lift force.

3 <u>**HOMEWORK**</u>

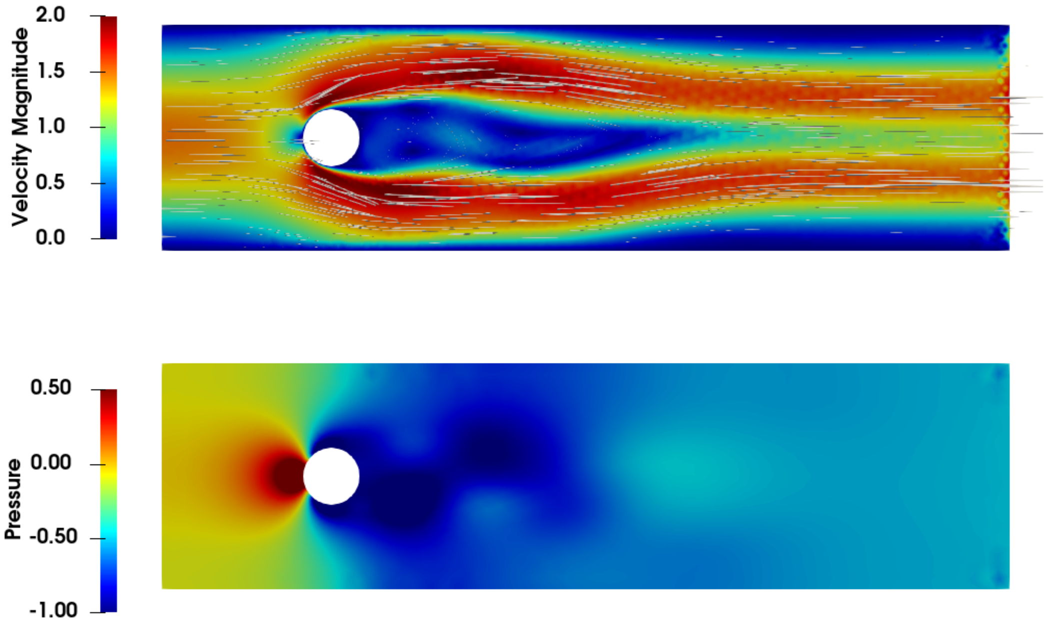

Figure 1: Velocity (top) and pressure (bottom) fields with no-slip conditions, at $t = 50$.

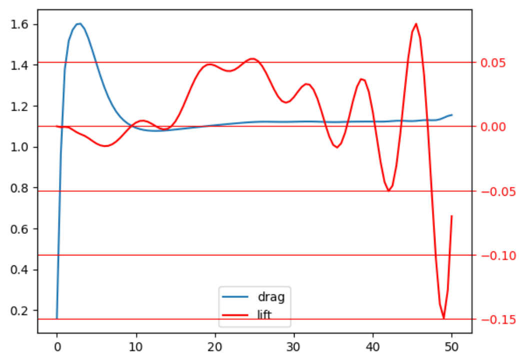

Figure 2: Time evolution of drag and lift forces on the obstacle for no-slip conditions.